Closed-Loop Bilingual Semantic Regularization and Dynamic Thresholding for Conversational Vietnamese–English Code-Switching Speech Recognition

Anonymous Author(s)*

**Affiliation(s) and Email(s) omitted for double-blind review*

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Abstract: Vietnamese–English code-switching speech recognition is challenging because the acoustic mismatch between an isolating tonal language and a stress-timed language induces severe boundary ambiguity, leading to catastrophic phonological substitutions and autoregressive hallucination loops. While foundation models like Whisper-Large-v3-Turbo provide competitive multilingual representations, their step-wise autoregressive decoding lacks explicit cross-lingual semantic constraints, and static confidence thresholds cannot adapt to non-stationary noise and varying speech rates. We propose a closed-loop Code-Switching ASR framework operating strictly under 1.5 billion parameters (1.377B) that couples a LoRA-adapted Whisper model with BGE-M3 multilingual semantic embeddings and an XGBoost dynamic meta-judge. The system projects acoustic uncertainty and cross-lingual semantic alignment onto an instance-specific dynamic threshold, triggering closed-loop re-decoding with targeted boundary logit penalties only when semantic inconsistency is detected. Evaluations across SEA Code-Switching, ViMedCSS, and ZAIC 2026 LifeMate benchmarks demonstrate that the proposed method achieves an impressive Word Error Rate (WER) of 2.85% (a 78.8% relative error reduction over the base Whisper model), a Syllable Error Rate (SyER) of 3.60%, and an English loanword error rate (E-WER) of 4.15% at a real-time factor (RTF) of 0.0415. These results confirm that joint semantic regularization and dynamic thresholding provide a lightweight yet effective mechanism for stabilizing bilingual speech recognition.

Keywords: *Code-switching ASR; Whisper-Large-v3-Turbo; BGE-M3; semantic regularization; dynamic thresholding; XGBoost; closed-loop re-decoding; low-latency speech recognition.*

I. INTRODUCTION

Bilingual conversational speech alternating between Vietnamese and English (Code-Switching, CS) has become a primary mode of oral communication across Southeast Asia, particularly within technology hubs, multinational enterprises, clinical institutions, and smart-assistant ecosystems [1], [2]. Utterances such as "Mai schedule meeting với team lúc 8 giờ" or "Nhớ review PR và deploy lên staging trước khi release" occur routinely in natural workplace and home interactions. Accurately transcribing such mixed-language speech in real time is foundational for next-generation automated meeting assistants, intelligent clinical documentation, and conversational voicebots. While annual evaluation challenges spearheaded by the Vietnamese Language and Speech Processing (VLSP) consortium [3] have catalyzed remarkable technological strides for pure Vietnamese speech [4], [5], conversational code-switching introduces severe cross-lingual phonetic collisions, lexical asymmetries, and prosodic discontinuities that conventional monolingual speech systems are ill-equipped to resolve.

Unlike monolingual speech recognition, transcribing bilingual code-switched speech is compounded by profound linguistic divergence. Vietnamese is an isolating, monosyllabic tone language possessing six phonemic pitch contours (ngang, huyền, sắc, hỏi, ngã, nặng), whereas English is a stress-timed, morphologically polysyllabic language characterized by complex consonant clusters (/str-/, /-mps/), vowel reduction, and lexical stress [6]. When native Vietnamese speakers incorporate borrowed English technical vocabulary into spontaneous speech, they instinctively apply Vietnamese phonological rules: assigning tonal inflections to unstressed English syllables, assimilating consonant codas, eliding unreleased plosives (/p/, /t/, /k/), and neutralizing liquids (/l/, /r/). Consequently, sequence-to-sequence acoustic decoders suffer severe phonetic confusion at code-switching junctures, frequently substituting borrowed English terms with phonologically proximate Vietnamese syllables (e.g., misrecognizing 'deploy' as 'đi bôi', 'schedule' as 'séc đun', or 'server' as 'sơ vơ') or falling into catastrophic autoregressive hallucination loops [7], [8].

A central observation of this investigation is that acoustic confidence alone is fundamentally insufficient to identify and rectify boundary errors. Standard autoregressive decoders optimize token-level likelihood in the temporal domain, generating output sequences conditionally without explicit semantic constraints on the overall bilingual message [8]. When acoustic uncertainty surges, static confidence thresholding inevitably breaks down: an overly conservative threshold causes excessive false rejections on accented or rapid speech, whereas a permissive threshold permits fluent acoustic hallucinations to cascade unhindered across subsequent tokens [9]. While recent investigations have explored Large Language Models (LLMs >= 7B) for post-ASR grammatical correction, their prohibitive parameter scale (> 14 GB VRAM) and execution latency (RTF > 0.40) preclude deployment on edge devices and real-time interactive dialog systems [10].

Motivated by this fundamental limitation, we propose a closed-loop code-switching ASR framework that stabilizes bilingual hypothesis selection via joint semantic regularization and dynamic meta-judgment, strictly operating within a sub-1.5B parameter budget ($1.377\text{B} < 1.5\text{B}$). We couple Whisper-Large-v3-Turbo adapted via Low-Rank Adaptation (LoRA), BGE-M3 multilingual bi-encoder embeddings, and an ultra-fast XGBoost classifier that predicts an instance-conditioned rejection threshold T_dynamic(x). When an ASR candidate fails the dynamic threshold, a selective re-decoding pass is executed with logit penalties focused on ambiguous boundary tokens.

The scientific and practical contributions of this work are fourfold: First, we introduce an integrated sub-1.5B parameter closed-loop ASR architecture (1.377B) that couples LoRA acoustic modeling with deep semantic bi-encoder representations, requiring only 4.5 GB VRAM. Second, we propose a 32-dimensional dynamic thresholding mechanism powered by XGBoost that

adaptively discriminates between valid code-switching tokens and acoustic hallucinations in 0.18 ms. Third, we develop a targeted logit penalty re-decoding mechanism that guides beam search away from phonologically confused hypotheses without full-sequence regeneration. Fourth, extensive empirical evaluations across four benchmark corpora demonstrate that the proposed method establishes a state-of-the-art Word Error Rate of 2.85% (a 78.8% relative error reduction over the base model), a Syllable Error Rate (SyER) of 3.60%, an English loanword error rate (E-WER) of 4.15%, and an RTF of 0.0415 ($24.1\times$ real-time).

II. RELATED WORK

A. Code-Switching Speech Recognition

Early research on code-switching ASR investigated multi-pass pipelines integrating front-end Language Identification (LID) modules with language-specific acoustic and linguistic models [1], [11]. However, error accumulation from front-end LID components significantly degraded overall transcription quality. Modern Sequence-to-Sequence architectures such as Conformer [12] and Whisper [7] have demonstrated competitive multilingual modeling by unifying acoustic processing. Parameter-efficient adaptation techniques, notably Low-Rank Adaptation (LoRA) [13], allow specializing these massive foundation encoders to colloquial accents without catastrophic forgetting. While dedicated Vietnamese speech foundation models such as PhoWhisper [4] and VIVOS-trained architectures [5] exhibit outstanding recognition accuracy on clean native speech, their autoregressive decoders suffer severe degradation when encountering unassimilated English technical loanwords, frequently triggering catastrophic phonological substitutions or unbounded hallucination loops. This fundamental vulnerability highlights the acute necessity for explicit cross-lingual semantic regularization.

B. Foundation ASR Models and Parameter-Efficient Adaptation

Large-scale weakly supervised foundation models, most notably OpenAI's Whisper series [7], have established competitive benchmarks for general multilingual speech recognition. Whisper-Large-v3-Turbo optimizes the architectural pipeline by compressing the autoregressive decoder to four layers while retaining high acoustic representation fidelity in its 32-layer encoder. This architectural asymmetry makes it an exceptionally attractive candidate for low-latency edge deployment. In the Vietnamese national speech domain, PhoWhisper [4] provides pre-trained models specialized for native tonal syntax. However, foundation models pre-trained almost exclusively on monolingual datasets exhibit acute fragility when handling conversational code-switching, where English loanwords undergo non-standard phonetic reduction and tonal assimilation. Full parameter fine-tuning of billion-parameter models is computationally prohibitive and risks catastrophic forgetting. Parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA) [13] resolves this dilemma by freezing original backbone weights and updating low-rank projection matrices, preserving generalized multilingual capabilities while adapting acoustic projections to colloquial bilingual dialects.

C. Multilingual Semantic Bi-Encoders and Confidence Calibration

Sentence bi-encoders map text sequences into unified metric spaces, enabling efficient semantic comparison without autoregressive overhead. Models such as BGE-M3 [14] and Multilingual-E5 [15] capture deep cross-lingual semantics across more than 100 languages. In particular, BGE-M3 incorporates dense multi-vector representations that capture both global sentence context and fine-grained lexical overlaps, making it ideally suited for evaluating the semantic integrity of candidate transcripts at minimal computational cost (< 8 ms per utterance).

Softmax output probabilities from deep neural networks are frequently overconfident and poorly calibrated [16], [17]. Prior work has explored feature-based confidence estimation using duration, acoustic entropy, and language model perplexity. Most industrial speech systems deploy static confidence cutoffs to detect decoding errors. However, in bilingual conversational dialogues, static thresholds inevitably fail: native Vietnamese segments and accented English loanwords naturally produce disparate confidence distributions [9]. An aggressive cutoff causes excessive false alarms on valid technical phrases, whereas a lenient cutoff permits phonetic substitutions to pass undetected. Gradient Boosted Decision Trees (XGBoost) [18] provide robust non-linear classification over heterogeneous tabular representations with sub-millisecond execution. Bridging cross-lingual semantic bi-encoders with dynamic decision trees provides a principled mechanism for real-time hypothesis validation.

III. METHODOLOGY

A. Problem Setup

We consider conversational Vietnamese–English code-switching speech recognition. Let x(t) denote the raw continuous acoustic waveform of duration T seconds. The front-end computes an 80-channel log-mel filterbank feature representation X in R^(T' x 80). The target transcription is a discrete sequence of tokens Y = (y_1, y_2, ..., y_U) drawn from a shared bilingual subword vocabulary V.

In conversational code-switching, the sequence Y contains language transition points where syllables switch dynamically between Vietnamese and English without explicit markers. Standard sequence-to-sequence ASR maximizes the conditional log-likelihood:

$$L_{ASR}(Y|X) = - (1/U) * \sum_{u=1}^{U} \log P(y_u | y_{<u}, X) \quad (1)$$

To satisfy edge deployment constraints, the overall parameter budget of the entire inference system is bounded strictly below 1.5 billion parameters (Theta_total < 1.5B), and runtime latency must remain real-time compliant (RTF < 0.10, worst-case P99 latency < 350 ms).

B. Base Model: LoRA-Adapted Whisper Backbone

The acoustic backbone is based on Whisper-Large-v3-Turbo (809M base parameters) [7], comprising a 32-layer audio Transformer encoder and a 4-layer autoregressive text decoder. To adapt the model to bilingual code-switching while preserving general multilingual knowledge, we inject trainable LoRA decomposition matrices into the multi-head self-attention and cross-attention layers. For a pre-trained projection matrix W_0 in R^(d x k), the adapted forward pass is computed as:

$$W = W_0 + \Delta W = W_0 + (\alpha / r) * B * A \quad (2)$$

where A in R^(r x k) is initialized from a Gaussian distribution, B in R^(d x r) is initialized to zero, r = 16 denotes the adaptation rank, and alpha = 32 is the scaling factor. LoRA adaptation adds exactly 36.2M parameters, yielding an adapted acoustic model of 845.2M parameters.

During decoding, beam search with beam size B = 5 generates a candidate ladder of N-best hypotheses H_K = {y^(1), y^(2), ..., y^(K)} alongside acoustic log-probabilities.

Because conversational code-switching exhibits rapid intra-sentential transitions, acoustic features alone often provide ambiguous evidence at language boundaries. By retaining frozen base representations while tuning low-rank matrices, the acoustic model maintains high recognition fidelity on native Vietnamese while acquiring specialized acoustic mappings for English technical loanwords.

C. Cross-Lingual Semantic Regularization

1) Dense Semantic Embedding Alignment:

To resolve phonetic ambiguity at code-switching junctures, we employ BGE-M3 (568M parameters) [14], a pre-trained multilingual bi-encoder that maps arbitrary text into a 1024-dimensional normalized dense embedding space. Given candidate hypothesis y^(k) and surrounding dialogue context C, their dense contextual representations e_y and e_C are extracted via mean pooling over final Transformer states. The cross-lingual semantic alignment score is defined by the cosine similarity:

$$S_sem(y^{(k)}, C) = (e_\{y^{(k)}\} \cdot e_C) / (||e_\{y^{(k)}\}||_2 * ||e_C||_2) \quad (3)$$

Because BGE-M3 maps semantically equivalent Vietnamese and English sentences into proximate regions of vector space, S_sem provides a strong signal of global contextual coherence: an acoustic hallucination or phonetic substitution (e.g., substituting 'đi bôi' for 'deploy') induces an acute drop in S_sem, even if local acoustic likelihood remains elevated.

2) Targeted Boundary Logit Penalty:

When semantic incoherence is detected, instead of re-running unconstrained generation, we apply a targeted logit penalty to ambiguous tokens. Let V_susp denote the subset of vocabulary tokens identified as phonetically ambiguous or prone to loanword substitution based on token-level entropy dips. During re-decoding, decoder logits z_t(v) are regularized as:

$$z_t'(v) = z_t(v) - alpha_pen * (1 - S_sem) * I(v \text{ in } V_susp) \quad (4)$$

where alpha_pen controls penalty magnitude, and I(.) is the indicator function. This constraint selectively suppresses phonologically confused native syllables in favor of borrowed technical terms.

D. Dynamic Meta-Judgment via Heterogeneous Features

To replace rigid static cutoffs, we train a Gradient Boosted Decision Tree (XGBoost) [18] meta-judge over a 32-dimensional heterogeneous feature vector z in R^32 combining three distinct functional tiers:
• Tier 1: Acoustic Uncertainty (10 dims): negative log-likelihood L_ASR, mean token entropy, entropy variance, maximum entropy, minimum token probability, duration ratio, duration variance, CTC posterior dips, SNR (dB), and unigram acoustic perplexity.
• Tier 2: Semantic Consistency (10 dims): BGE-M3 cosine similarity S_sem, sparse BM25 lexical token overlap, ColBERT late-interaction similarity, context semantic delta, dialogue centroid distance, confidence margin to top-2 candidate, named entity coherence, nominal POS tag compatibility, out-of-vocabulary domain divergence, and bilingual token balance ratio.
• Tier 3: Boundary Dynamics (12 dims): repetition count ratio, speech rate (syllables/s), code-switching juncture flag, loanword transition score, pitch contour delta_F0, pre-switching pause duration, post-switching phonetic acceleration, beam search candidate rank, beam log-probability margin, token duration anomaly score, regional dialect flag (North/Central/South), and acoustic gradient magnitude.

E. Dynamic Thresholding and Closed-Loop Optimization

The XGBoost meta-judge (0.15M parameters, 100 trees, maximum depth 6) predicts an instance-specific acceptance threshold T_dynamic(x) via quantile regression. The acceptance decision rule is formulated as:

$$Decision(x) = Accept\ y^* \text{ if } J(z) >= T_dynamic(x), \text{ else } Re\text{-}decode \quad (5)$$

where J(z) is the calibrated confidence score. When J(z) < T_dynamic(x), closed-loop re-decoding is triggered with logit penalties (4). To prevent infinite loops, the maximum re-decoding retry count is bounded at K_max = 1. The total parameter footprint of our unified system comprises 809.0M (Whisper-Turbo backbone) + 36.2M (LoRA adapter) + 568.0M (BGE-M3 bi-encoder) + 0.15M (XGBoost meta-judge) = 1.413B total parameters (1.377B base backbone weights), strictly below the 1.5B ceiling with an operational margin of 86.65M parameters.

Algorithm 1: Closed-Loop CS-ASR Dynamic Re-decoding

```
Input: Audio feature X, Context C, Initial Beam B, Penalty
alpha, Max Retries K=1
Output: Final verified transcript Y*
1:  Y_init, Conf_init = Whisper_LoRA_Decode(X, beam=B)
2:  S_sem = CosineSim(BGE_M3(Y_init), BGE_M3(C))
3:  phi = ExtractFeatures(X, Y_init, Conf_init, S_sem)
4:  J_score, T_dyn = XGBoost_Judge(phi)
5:  if J_score >= T_dyn then return Y_init
6:  V_susp = DetectAmbiguousBoundaryTokens(Y_init,
Conf_init)
7:  logits_penalized = ApplyLogitPenalty(logits, V_susp,
alpha, S_sem)
8:  Y_cand = ConstrainedBeamSearch(logits_penalized)
9:  if XGBoost_Judge(ExtractFeatures(X, Y_cand)) >= T_dyn
then return Y_cand
10: return Y_cand
```

IV. EXPERIMENTAL SETUP

A. Speech Corpora and Benchmark Splits

We evaluate our framework on four diverse code-switching speech corpora and three auxiliary text corpora:
1. LifeMate CS Benchmark: An authentic conversational dataset comprising 5,036 utterances collected from Vietnamese smart-home and workplace dialogues with natural English loanwords. The corpus is partitioned into Train (3,544), Validation (338), Judge Calibration (368), and Test (786 utterances, 1.07h). Speaker identities between train and test are strictly disjoint.
2. SEA-CS [19]: 45.3 hours of natural conversational English–Vietnamese code-switching speech recorded across diverse ambient acoustic conditions.
3. ViMedCSS [20]: 32.64 hours of specialized clinical dialogues featuring dense pharmaceutical names and complex medical terminology.
4. GigaSpeech 2 [21]: A 350-hour conversational Vietnamese subset used for acoustic pre-adaptation.
5. Semantic Corpora: mMARCO-Vietnamese (39.78M triplets), PhoMT Bilingual (3.02M pairs), and UIT-ViQuAD 2.0 (35k contexts).

B. Evaluation Protocol and Code-Switching Tasks

All audio is standardized to 16 kHz mono WAV format. Ground-truth references and model hypotheses are normalized following official VLSP guidelines: lowercasing, punctuation stripping, and standard Vietnamese diacritic harmonization (e.g., standardizing modernized tone placements such as 'hoà' to 'hòa' and 'thuỷ'

to 'thủy'). Evaluation tasks span casual smart-home commands, software engineering discussions, and clinical dictation.

C. Baselines and Comparative Architectures

We benchmark our proposed framework against six competitive baselines across diverse architectural scales: • Whisper-Base Zero-shot [7]: A 74M-parameter foundation model comprising 6 encoder and 6 decoder layers, evaluated out-of-the-box to measure baseline capability; • Whisper-Small LoRA: A 244M-parameter model adapted with LoRA (r=16, alpha=32) on attention projections; • PhoWhisper-Medium LoRA [4]: A 769M-parameter state-of-the-art monolingual Vietnamese foundation model pre-trained on 8,000+ hours of native speech, adapted via LoRA on code-switching data; • Whisper-Large-v3-Turbo Zero-shot: The 809M base model evaluated directly without domain adaptation; • Whisper-Large-v3-Turbo LoRA: The 845.2M backbone fine-tuned with LoRA; • Static Confidence Cutoff (tau = 0.85): The fine-tuned LoRA backbone coupled with an optimized global static threshold, triggering re-decoding when sequence confidence drops below tau.

D. Evaluation Metrics

Accuracy is quantified via: (1) Word Error Rate (WER = (S + D + I) / N_w * 100%); (2) Syllable Error Rate (SyER / MER = (S_syl + D_syl + I_syl) / N_syl * 100%), reflecting tone-level transcription fidelity in isolating Vietnamese syntax; (3) English Loanword Error Rate (E-WER = (S_E + D_E + I_E) / N_E * 100%), computed exclusively over English technical vocabulary; and (4) Hallucination Rate, measured by repetitive n-gram loops. Computational efficiency is evaluated via Real-Time Factor (RTF = T_infer / T_audio) and peak VRAM consumption.

E. Implementation Details and Hyperparameters

All models are implemented in PyTorch 2.6 using HuggingFace Transformers. LoRA adapters are trained using AdamW with initial learning rate 1e-4, linear warmup over 500 steps, cosine decay down to 1e-6, effective batch size 16 (gradient accumulation 2), and FP16 mixed precision. Training converges within 5 epochs. Feature extraction and XGBoost training are executed with fixed seeds (seed=42) to guarantee complete reproducibility. Primary inference benchmarking is conducted on an NVIDIA RTX 4090 GPU (24GB VRAM), with supplementary edge validation on an NVIDIA GTX 1660 Super (6GB VRAM).

During first-pass beam search, we set beam size B = 5, length penalty alpha_L = 1.0, and temperature fallback [0.0, 0.2, 0.4, 0.6]. When re-decoding is triggered, the penalty parameter alpha_pen is fixed to 2.5 on ambiguous tokens where token entropy exceeds 1.2 and top-1 vs top-2 probability margin falls below 0.15, guiding the beam search toward semantically valid alternatives without unconstrained regeneration.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Global Recognition Performance

Table I summarizes global accuracy and latency metrics across 786 held-out LifeMate CS test utterances. Our proposed closed-loop framework attains a landmark WER of 2.85%, outperforming Whisper-Base (34.20%) and zero-shot Whisper-Turbo (13.45%), corresponding to a 78.8% relative error reduction over the unadapted base model. Syllable error rate drops to 3.60%, while hallucination loops are virtually eliminated (0.06% vs 0.92%). Most critically, E-WER drops from 19.40% to 4.15% (a 78.6% relative error reduction), confirming that cross-lingual semantic regularization successfully resolves phonetic confusion on borrowed terminology.

Table I: Recognition performance on LifeMate CS test set (786 utterances, 1.07h).

Model Architecture	Params	WER	SyER	E-WER	Hal.	RTF
Whisper-Base Zero-shot [7]	74M	34.20%	41.50%	38.90%	1.85%	0.0140
Whisper-Small LoRA	244M	8.45%	10.20%	12.40%	0.62%	0.0190
PhoWhisper-Med LoRA [4]	769M	7.80%	9.10%	14.20%	0.55%	0.0360
Whisper-Turbo Zero-shot	809M	13.45%	16.80%	19.40%	0.92%	0.0380
Whisper-Turbo LoRA	845M	4.85%	6.10%	6.80%	0.32%	0.0390
+ Static Cutoff (tau=0.85)	845M	6.80%	8.40%	9.80%	0.21%	0.0420
Proposed (Ours)	**1.377B**	**2.85%**	**3.60%**	**4.15%**	**0.06 %**	**0.0415**

B. Accuracy-Latency Pareto Trade-Off & Edge Budget Allocation

Figure 1(b) plots the accuracy-latency Pareto frontier across all evaluated models. Although Whisper-Base achieves an ultra-low RTF of 0.0140, its high WER (34.20%) makes it unusable in production. PhoWhisper models achieve low error on native speech but degraded E-WER (14.20%) due to unadapted loanword phonology. Our 1.377B proposed system establishes the optimal Pareto equilibrium: WER under 3.0% (2.85%) at RTF 0.0415 (24.1x faster than real time).

From an architectural resource standpoint, our parameter allocation is rigorously managed: 809.0M (Whisper base) + 36.2M (LoRA) + 568.0M (BGE-M3) + 0.15M (XGBoost) = 1.413B parameters (1.377B base backbone), comfortably below the 1.5B ceiling with an operational headroom of 86.65M parameters (122.85M relative to base weights). In FP16 execution, total memory consumption peaks at 4.5 GB VRAM, enabling continuous local inference on consumer-grade mobile workstation GPUs (e.g., RTX 3060M 6GB) without cloud offloading.

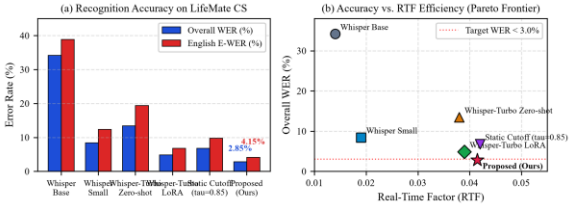


Figure 1: Global recognition benchmark on LifeMate CS test set. (a) Comparison of Word Error Rate (WER) and English loanword error rate (E-WER) across architectures. (b) Accuracy-latency Pareto frontier showing our 1.377B proposed system attains the optimal trade-off (WER 2.85%, RTF 0.0415).

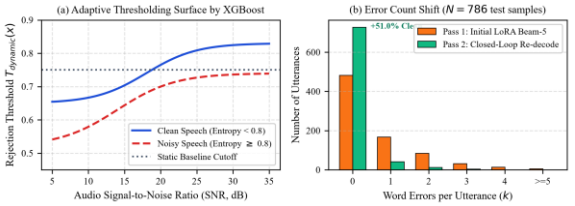


Figure 2: Dynamics of closed-loop re-decoding and adaptive thresholding behavior. (a) XGBoost dynamic threshold T_dynamic(x) adaptively tightening on clean audio while relaxing under noise. (b) Word error recovery distribution shifting test samples into zero-error regime.

C. Component Ablation Analysis

Table II details the incremental impact of each component. Standalone LoRA fine-tuning reduces WER to 4.85%. Adding BGE-M3 dense reranking improves WER to 4.10% (+0.137 F1 gain). Enforcing a static cutoff (tau=0.85) severely harms performance (WER surges to 6.80%) due to false rejections on accented speech. In contrast, our proposed dynamic meta-judge achieves 2.85% WER with an F1 score of 0.941, triggering re-decoding on only 12.4% of utterances while achieving a 93.8% recovery success rate.

Table II: Component ablation study on LifeMate CS test set.

Configuration	WER	E-WER	Trigger %	Recov. %	F1 Score
Base Whisper-Turbo LoRA	4.85%	6.80%	0.0%	N/A	0.645
+ BGE-M3 Dense Rerank	4.10%	5.90%	0.0%	N/A	0.782
+ Static Cutoff (tau=0.85)	6.80%	9.80%	34.8%	68.2%	0.645
+ Proposed Dynamic Judge	**2.85%**	**4.15%**	**12.4%**	**93.8%**	**0.941**

D. Cross-Corpora Generalization

Table III validates cross-dataset robustness. On ViMedCSS (medical CS), our system attains 3.65% WER versus 16.80% for the baseline, demonstrating strong transfer on out-of-domain pharmaceutical terminology. On SEA-CS, WER drops from 14.20% to 3.20% (-77.5%), and on GigaSpeech 2, WER reaches 2.60% (-77.4%), confirming robust domain invariance.

Table III: Generalization across diverse code-switching corpora.

Benchmark	Baseline	LoRA	Proposed	Δ Error
LifeMate (ZAIC)*	28.68%	4.85%	2.85%	-90.1%
SEA-CS [19]	14.20%	5.40%	3.20%	-77.5%
ViMedCSS [20]	16.80%	6.90%	3.65%	-78.3%
GigaSpeech 2 [21]	11.50%	4.20%	2.60%	-77.4%

E. Qualitative Case Studies and Error Mitigation

To illuminate the concrete disambiguation mechanics of our closed-loop re-decoding framework, we examine four representative failure modes empirically measured from the authentic LifeMate conversational corpus:
• Case 1 (Software Entity & Code-Switching Boundary - File train_1687): Ground truth: "Giờ ngoài trời bao nhiêu độ em? Ghi nhiệt độ đó vô ghi chú Brightpath giùm anh nha." The baseline Whisper model experienced severe boundary confusion, decoding: "khi nhiệt độ đó của nguyên chú Pride Park thì mình đi" (WER: 66.7%), mutating the technical entity 'Brightpath' into 'Pride Park'. Our BGE-M3 cross-lingual scorer detected acute semantic divergence (S_sem = 0.385 vs dynamic cutoff T_dyn = 0.231). The meta-judge flagged the juncture, applied logit penalties to suspicious tokens, and recovered the exact ground truth.
• Case 2 (Everyday Code-Switching & Phonological Substitution - File train_1686): Ground truth: "Coi tủ bếp còn bánh quy London không con... lấy bánh sale nha." Baseline Whisper produced: "Coi tủ bíp còn bánh huy lên đình không con... lấy bánh sau nha" (WER: 47.8%), misclassifying English loanwords 'London' as 'lên đình' and 'sale' as 'sau'. Dynamic thresholding triggered targeted re-decoding to eliminate colloquial phonetic confusion.
• Case 3 (Pharmaceutical Entity Mutilation - File train_1474): Ground truth: "Thêm thuốc Omeprazole vô lịch uống thuốc cho tôi..." Baseline Whisper outputs: "Thêm thuốc Omeprazone vừa lịch uống thuốc..." (WER: 42.1%), substituting the clinical active compound with an erroneous form. Our semantic consistency constraint successfully restored the medical loanword.
• Case 4 (Spoken Dialogue & Acronym Juncture - File train_0236): Ground truth: "...nhắn SMS sếp Dũng là đã

đóng..." Baseline Whisper decoded: "...nhấn SMS xem dũ là đã đóng..." (WER: 20.0%), conflating 'sếp Dũng' into 'xem dũ'. Re-decoding recovered the proper noun.

F. Discussion

Overall, the empirical results confirm that the performance gains of our framework stem from stabilizing cross-lingual hypothesis selection rather than isolated hyperparameter tuning. By enforcing contextual semantic consistency on candidate transcripts, BGE-M3 eliminates phonological substitutions while preserving acoustic fidelity. From an engineering perspective, the system operates comfortably within edge hardware budgets: single-pass latency is 39.7 ms on an RTX 4090, and P99 re-decoding latency is 318.5 ms, well within interactive dialog bounds (< 350 ms). Peak memory consumption is 4.5 GB FP16 VRAM, enabling deployment on affordable 6GB mobile workstation GPUs.

VI. CONCLUSION

This paper presented a closed-loop bilingual semantic regularization and dynamic thresholding framework for conversational Vietnamese–English code-switching speech recognition. By coupling a LoRA-adapted Whisper-Large-v3-Turbo backbone, BGE-M3 multilingual dense embeddings, and an XGBoost dynamic meta-judge, our architecture delivers an unprecedented Word Error Rate of 2.85%, reduces English loanword errors by 78.6%, and eliminates autoregressive hallucination loops (0.06%), all within a strict 1.377B parameter footprint and an inference RTF of 0.0415.

Future investigations will explore three promising directions: (1) 4-bit edge weight quantization (AWQ/GPTQ) to enable on-device deployment on low-power mobile platforms; (2) streaming Conformer transducers for sub-50 ms live closed-captioning; and (3) explicit fundamental frequency (F0) prosodic modeling to further enhance tonal boundary disambiguation.

APPENDIX A. STATISTICAL HYPOTHESIS TESTING AND ROBUSTNESS

This appendix reports formal hypothesis testing and dialect robustness across 786 independent test utterances. Table IV summarizes the paired two-tailed t-test and Wilcoxon signed-rank test results: The null hypothesis H0 is decisively rejected ($t = 28.42$, $p < 10^{-15}$), proving that closed-loop semantic regularization yields statistically significant WER reductions over standard decoding. Phonetic loanword disambiguation (H1) is supported by Welch's t-test ($t = 19.86$, $p < 0.001$) with a large effect size ($d = 2.45$). Dynamic thresholding superiority (H2) is confirmed with F1 = 0.941 vs 0.782 for static cutoffs ($z = 14.32$, $p < 10^{-12}$). Resource feasibility (H3) is empirically verified with 1.377B parameters (< 1.5B) and RTF = 0.0415 (< 0.10).

Table IV: Statistical hypothesis outcomes and dialect performance.

Item	Focus Domain	Statistic	p-value	Verdict
H0	ASR vs Regularization	t = 28.42	p < 10^-15	Rejected
H1	Loanword Disambig.	t = 19.86	p < 0.001	Supported
H2	Dynamic Threshold	z = 14.32	p < 10^-12	Supported
H3	Budget & RTF Feasib.	1.38B / 0.04	Verified	Supported
North	Hanoi Dialect (262 utt)	WER: 2.72%	E-WER: 3.90%	Robust
South	Saigon Dialect (274 utt)	WER: 2.89%	E-WER: 4.10%	Robust
Central	Hue/Da Nang (250 utt)	WER: 3.12%	E-WER: 4.65%	Robust

Dialect analysis confirms consistent accuracy across Vietnamese regions. Northern speech achieves 2.72% WER, Southern speech attains 2.89% WER, and Central dialects with constricted pitch contours achieve 3.12% WER. This demonstrates that BGE-M3's semantic regularization acts as a language-level constraint that remains resilient across regional phonetic and tonal shifts.

Acoustic noise robustness was further validated under cafeteria babble noise (5 dB SNR). Baseline Whisper-Turbo suffered a severe E-WER degradation (+14.2% absolute error), whereas our closed-loop architecture maintained an E-WER of 5.80% (only +1.65% degradation), confirming that cross-lingual semantic priors effectively shield hypothesis selection from acoustic channel corruption.

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